



Community Composting 101



The What & Why of Composting

Neighborhood
Soil Rebuilders
COMPOSTER TRAINING PROGRAM

ILSR
INSTITUTE FOR
Local Self-Reliance

The *What* & *Why* of Composting

- ✓ Define key composting related terms
- ✓ Introduce the composting process
- ✓ Discuss benefits of composting
- ✓ Where community composting fits into the hierarchy of food waste recovery strategies

What is Composting?

An illustration of a garden scene. The top half shows green plants with red and orange berries, a yellow sunflower, and a yellow and black butterfly. The bottom half shows a cross-section of brown soil with the roots of the plants extending downwards.

What is Composting?

A circular inset with a yellow border showing a pile of food scraps on a wooden surface. The scraps include two cracked eggs, apple cores, carrot peels, green herbs, and other vegetable waste.

*Photo:
Jessica Hamm,
Rochester, NY*



What is Composting?

Photo: ILSR



What is Composting?

- ▶ The controlled decomposition of raw **organic materials**
- ▶ An **aerobic** process, requiring oxygen
- ▶ Driven by several cycles of organisms that feed on organic materials
- ▶ The creation of a stable, soil enhancing product known as **compost**.



Source: ILSR



What is Composting?

The Red Hook Community Farm Compost Operation in Brooklyn, NY

Photo: @RedHookCompost



@RedHookCompost



ILSR Recreates D.L. Dindal's original illustration from 1978.

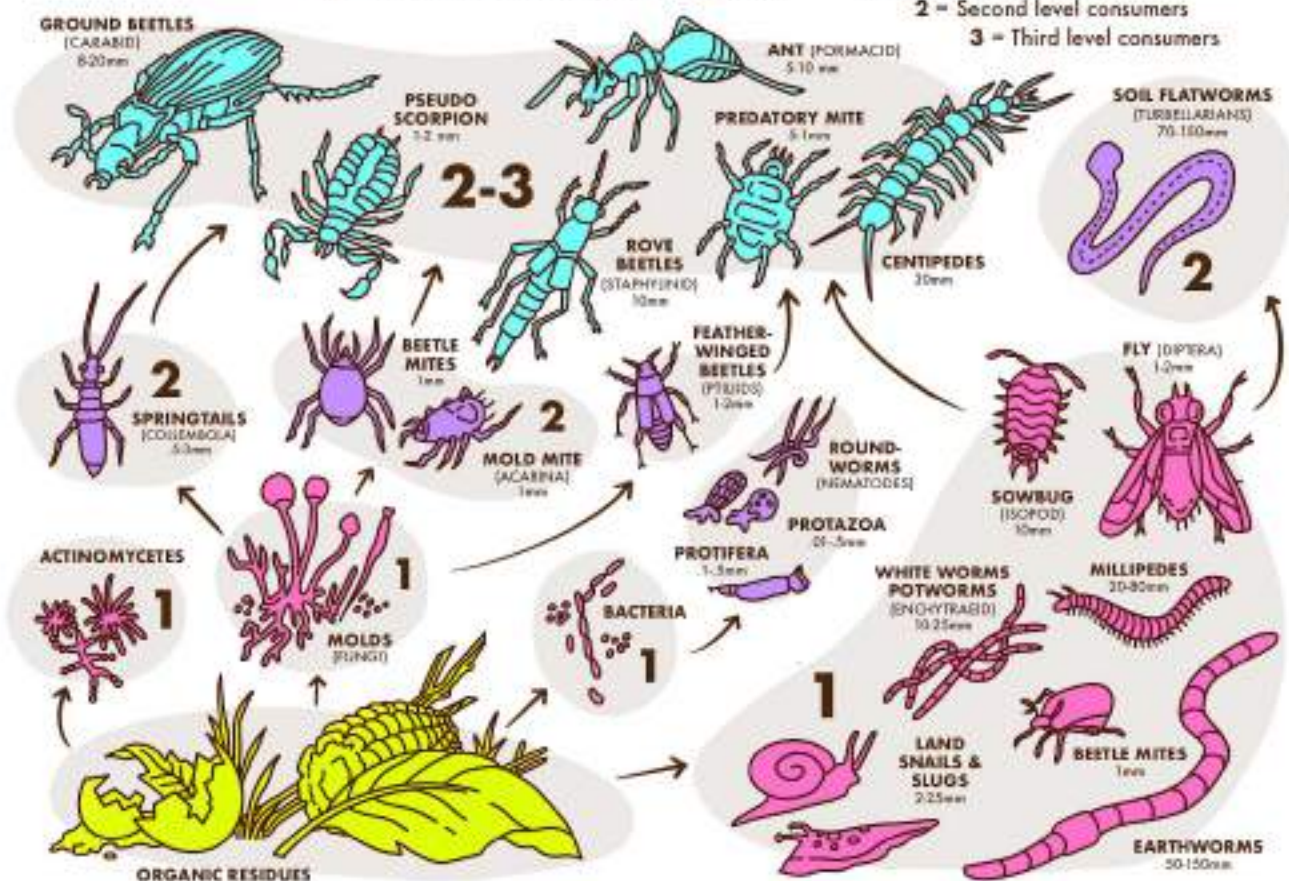
THE COMPOST PILE FOOD WEB

The energy flows in the direction of the arrow.

1 = First level consumers

2 = Second level consumers

3 = Third level consumers





What Is Compost?

- ▶ A dark, crumbly, earthy-smelling product of the decomposition of *organic materials*
- ▶ A living soil amendment that:
 - Improves soil's physical, biological & chemical properties
 - Enhances soil's ability to support plants & beneficial organisms



Photo: ILSR

Why Compost?

An illustration of a garden scene. At the top, there are green plants with red and orange berries, a yellow sunflower, and a yellow and black butterfly. Below the plants is a layer of dark brown soil. The title 'Why Compost?' is written in large white letters across the top.

WHY COMPOST?

1 TO **REDUCE** YOUR **WASTE**

> 50% of
municipal
garbage is
compostable

You can compost: food
scraps, yard trimmings,
paper, wood waste
& more!

COMPOST:

increases
soil fertility &
suppresses
plant disease

improves
soil
structure

increases
microbial activity
& water retention
capacity

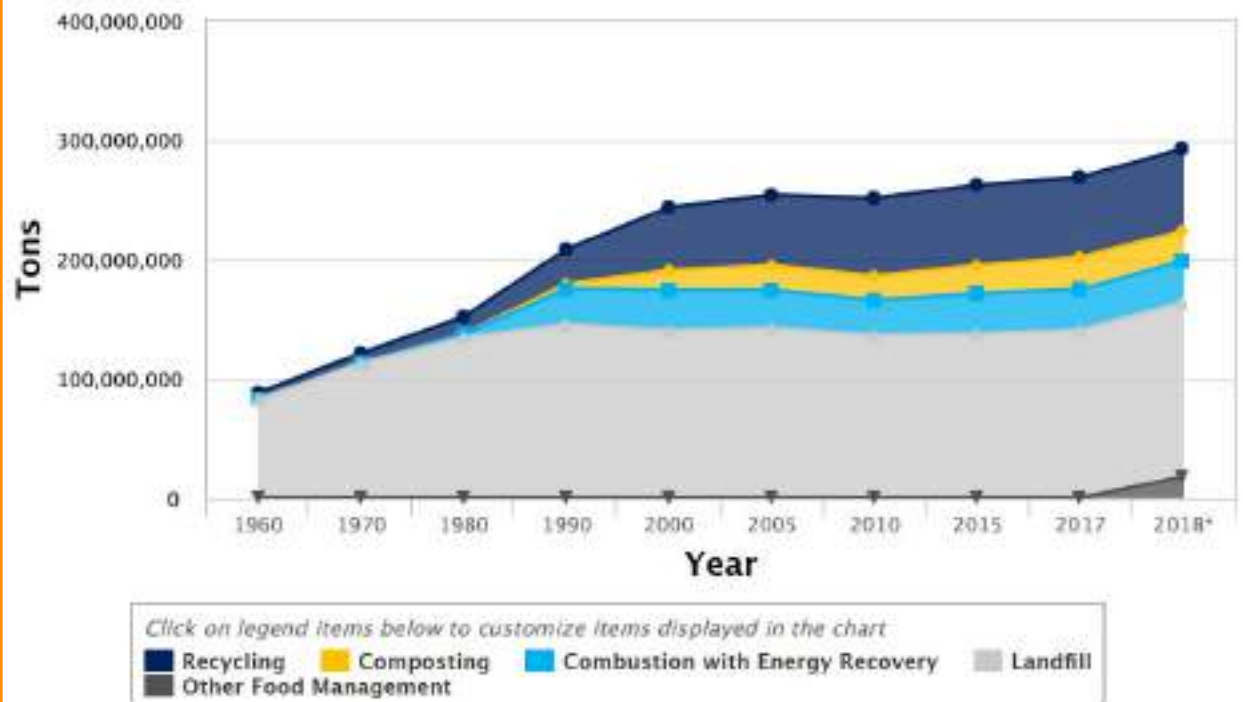
2 TO **ENHANCE** YOUR **SOIL**



Data Source: "National Overview: Facts and Figures on Materials, Wastes and Recycling," US EPA, 2018
Graphic: ILSR



Municipal Solid Waste Management: 1960 - 2018



Source: "National Overview: Facts and Figures on Materials, Wastes and Recycling," US EPA, 2018.

Wasted Food: 1960 - 2018



Source: "National Overview: Facts and Figures on Materials, Wastes and Recycling," US EPA, 2018.



AUGUST 2017
A17-004

REPORT

WASTED:

HOW AMERICA IS LOSING UP TO 40 PERCENT OF ITS FOOD FROM FARM TO FORK TO LANDFILL

SECOND EDITION OF NRDC'S ORIGINAL 2012 REPORT



<https://www.nrdc.org/resources/wasted-how-america-losing-40-percent-its-food-farm-fork-landfill>



MORE THAN JUST FOOD

THE U.S. WASTES TONS OF RESOURCES WHEN WE WASTE FOOD

1,250 CALORIES PER PERSON PER DAY
THAT IS HALF OF THE RECOMMENDED DAILY INTAKE FOR ADULTS

19%
OF ALL
U.S.
CROPLANDS
THAT IS MORE
LAND THAN ALL
OF NEW MEXICO

21% OF U.S. LANDFILL
CONTENT

THE NO. 1 CONTRIBUTOR BY WEIGHT

18%
OF ALL
FARMING
FERTILIZER
WHICH CONTAINS
3.9 BILLION POUNDS
OF NUTRIENTS

2.6% OF ALL U.S. GREENHOUSE
GAS EMISSIONS ANNUALLY


27 MILLION PASSENGER VEHICLES' WORTH

21% OF THE U.S. AGRICULTURAL
WATER USAGE




MORE THAN: TEXAS + CALIFORNIA + OHIO

\$218,000,000,000
WHICH IS EQUAL TO 1.3% OF THE U.S. GROSS DOMESTIC PRODUCT (GDP)

Source: *Wasted: How America is Losing Up to 40 Percent of Its Food From Field to Fork to Landfill*, Natural Resources Defense Council, 2017.



Reducing Wasted Food At Home

Most people don't realize how much food they throw away every day — from uneaten leftovers to spoiled produce. [EPA estimates that in 2010](#), about 68 percent of the wasted food we generated—or about 42.8 million tons— ended up in landfills or combustion facilities. By managing food sustainably and reducing waste, we can help businesses and consumers save money, provide a bridge in our communities for those who do not have enough to eat, and conserve resources for future generations.

On this page:

- [Benefits of Reducing Wasted Food](#)
- [Ways to Reduce Wasted Food](#)
 - [Planning Tips](#)
 - [Storage Tips](#)
 - [Prep Tips](#)



Source: "Reducing Wasted Food at Home," US EPA.



Source: "274,000 Pounds of Food (and Counting) Recovered," Capitol News Service, 2019.
Image: Jason Gembicki

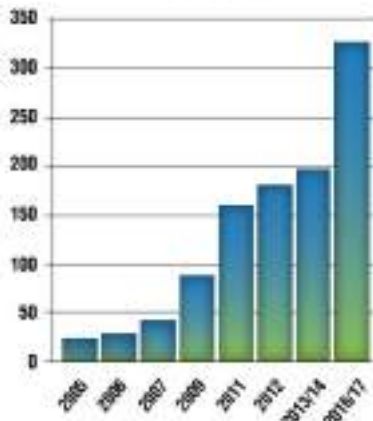


Source: "Food Scraps for Pigs," Baltimore Sun, 2019.
Image: Kenneth K. Lam

How much is the U.S. composting?

U.S. communities with curbside food waste collection

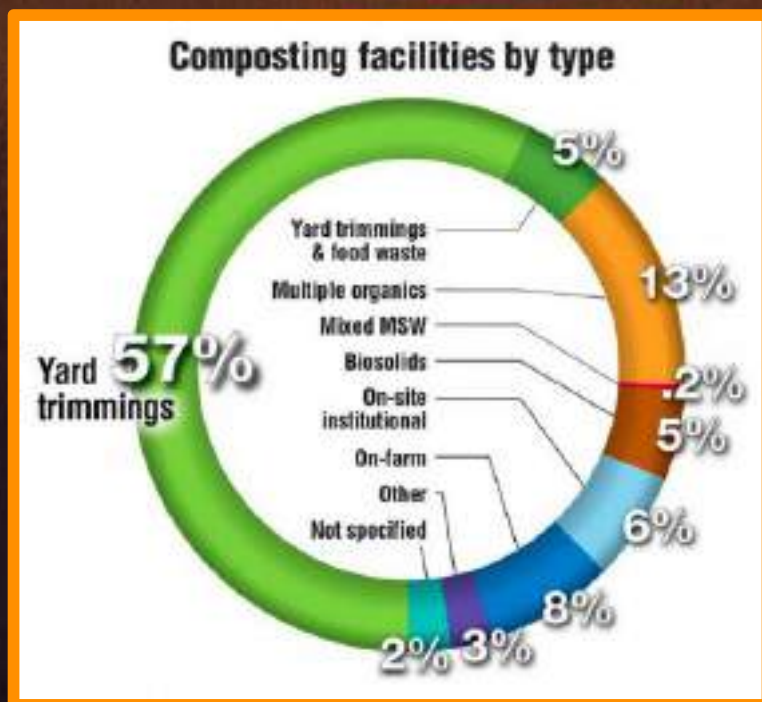
148 Programs reporting



Source: Virginia Streeter & Brenda Platt (ILSR), "Residential Food Waste Collection Access In The U.S.," BioCycle, December 6, 2017.



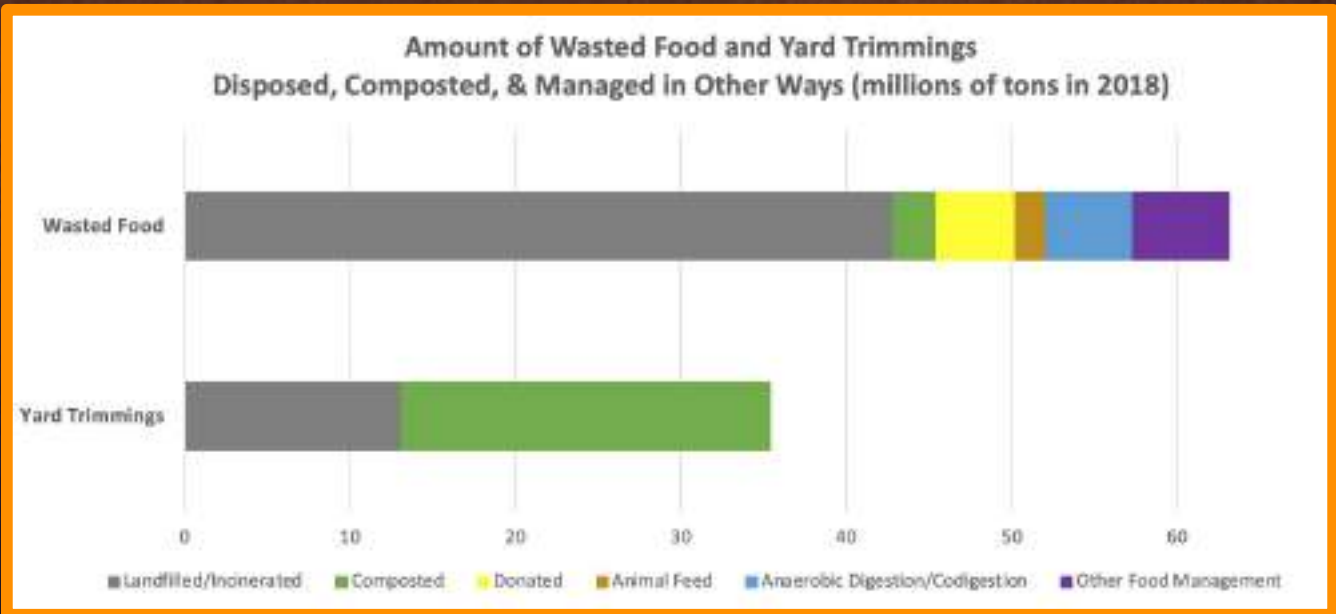
How much is the U.S. composting?



Source: Nora Goldstein, "The State of Organics Recycling In The U.S.," BioCycle, October 4, 2017.



How much is the U.S. composting?



Source: Institute for Local Self-Reliance chart based on
"National Overview: Facts and Figures on Materials, Wastes
and Recycling," US EPA, 2018



Composting: lots of ways & sizes

Backyard in
Takoma Park, MD



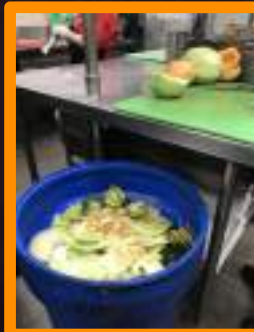
Prospect Heights
Community Garden, Brooklyn, NYC



Rippowam Cisqua School, Bedford, NY
Source: Rippowam Cisqua School website



University of
Maryland
Research Farm
(CMREC)



St. John's University,
Queens, NYC



Hierarchy to Reduce Food Waste and Grow Community



Hierarchy to Reduce Food Waste and Grow Community



Prevention. Do not generate food waste in the first place! Reduce portions, buy what you need, and organize your fridge for optimal food usage.

Hierarchy to Reduce Food Waste and Grow Community



Feed hungry people. Divert food not suitable for people to animals such as backyard chickens or to local farmers' livestock.

Hierarchy to Reduce Food Waste and Grow Community



Composting in backyards or in homes. Avoid collection costs!

Hierarchy to Reduce Food Waste and Grow Community



Onsite composting or anaerobic digestion, and community composters can accept material from off-site or simply process their own material.

Hierarchy to Reduce Food Waste and Grow Community



Composting or anaerobic digestion at the small town or farm scale. These systems handle typically between 10 and 100 tons per week and are designed to serve small geographic areas.

Hierarchy to Reduce Food Waste and Grow Community



Facilities serving large geographic areas that typically handle more than 100 tons per week. Material generally leaves the community in which it is generated.

Hierarchy to Reduce Food Waste and Grow Community



Mixed garbage is mechanically and biologically processed to recover recyclables and reduce waste volume and the potential for methane emissions before landfill disposal.

Hierarchy to Reduce Food Waste and Grow Community



Food waste should be banned from landfills and trash incinerators due to their high capital costs, pollution, and contribution to greenhouse gas emissions.



Composting Enhances Soil and Protects Watersheds

Healthy soils are essential for protecting watersheds. Compost is the best way to add organic matter—which is vital—to soils.

When added to soil, compost can filter out urban stormwater pollutants by an astounding **60-95%**

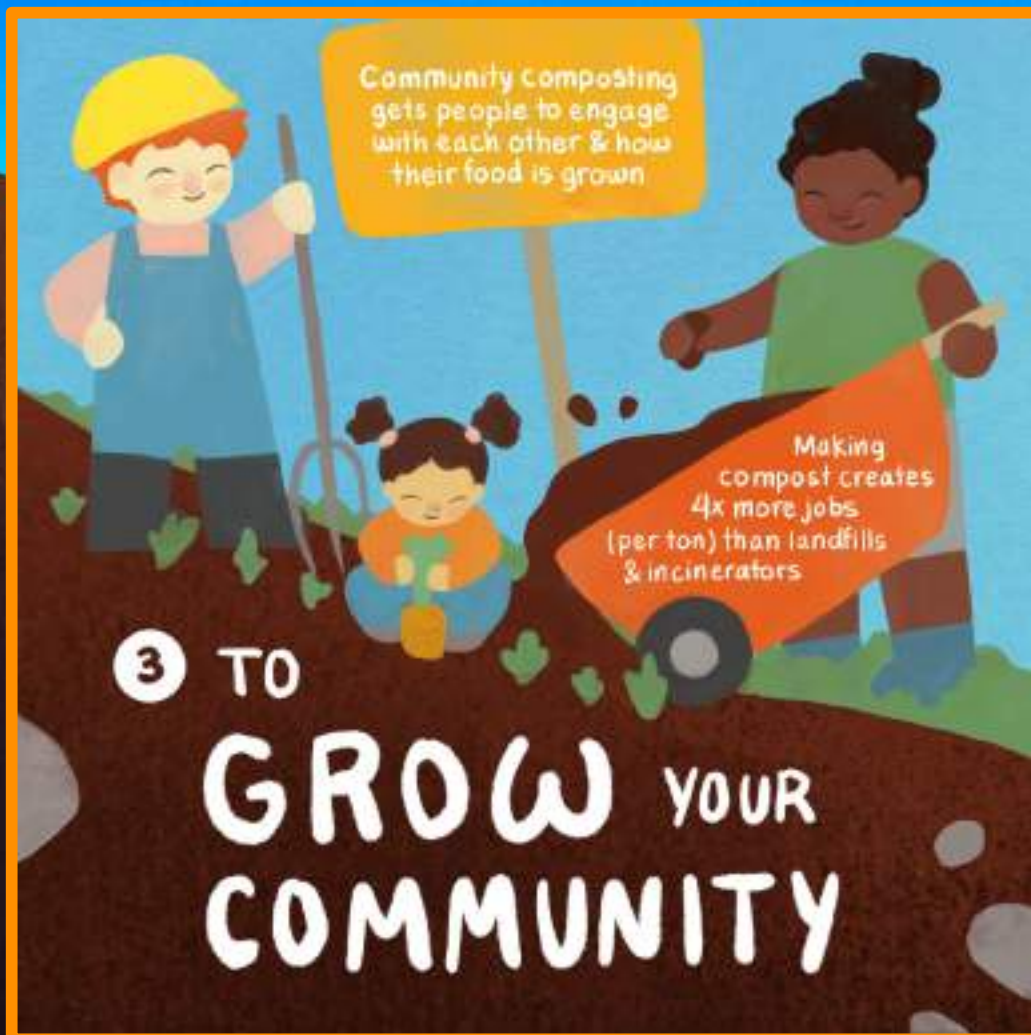


Compost & Organic Matter

- ▶ **Organic matter** is the product of the natural decomposition of raw **organic materials**
- ▶ **Compost** is the product of the controlled decomposition of raw **organic materials**

"THERE IS NO
BETTER PLANT
FOOD THAN
THE THINGS
WE IGNORE OR
THROW AWAY
EVERYDAY."

• GEORGE
WASHINGTON
CARVER





VS.



ECO City Farms

Photo: ILSR



Creating a Circular Food System

“The linear food system is ripe for disruption. For every dollar spent on food, society pays two dollars in health, environmental, and economic costs. With 80% of all expected to be consumed in cities by 2050, businesses, public bodies, organizations, institutions, and the people located within them hold the power to revolutionize our food system.”

Source: *Cities & The Circular Economy for Food*,
Ellen McArthur Foundation, 2019.



Creating a Circular Food System

COMPOSTING IS NOT A LUXURY. IT IS AN ACT OF PROTEST, A MODE OF RESISTANCE, AND A WAY TO ACTIVELY FIGHT BACK AGAINST ENVIRONMENTAL INJUSTICE.

Source: Jesi Taylor Cruz @moontwerk



WHY COMPOST?

FEED THE SOIL THAT FEEDS YOU



Source:
Baltimore Office of
Sustainability's Food
Matters Program



Closing the Loop with Compost



ECO City Farms in Prince George's County, MD.

Photos: ECO City Farms



Composting Creates Jobs

Jobs are sustained in each stage of the organics recovery cycle.

PER 10,000 TONS WASTE/YEAR

JOB SUSTAINED



Green infrastructure uses compost in rain gardens, green roofs, bioretention, vegetated retaining walls, and on steep highway embankments to control soil erosion and storm water. Using compost in green infrastructure creates **even more jobs**.



On a per-ton basis, making compost alone, employs 2x more workers than landfills and 4x more than incinerators.





Denbow, www.denbow.com



MCS Inc., www.mcsnjinc.com



Community Composters Can Do This Too!



COMPOST FILTER SOCK

Compost Filter Sock is a durable woven fabric filled with composted manure that is designed to redirect, flow and filter out sediment and other contaminants from stormwater events using a three-dimensional filtration process.

WHY USE COMPOST FILTER SOCK?

Compost Filter Sock is a cost-effective BMP that saves money over the life of a construction project due to ease of installation, maintenance, and disposal.

Compost Filter Sock slows the flow while still allowing water to filter through. This prevents large pooling or flooding which saves time and money lost cleaning up muddy sidewalks and streets.

Avoid paying minimum install fees using your own crew to install when and where you want it.



Fertile Ground Cooperative, Oklahoma City:
<https://www.fertilegroundok.coop/compostfiltersock>



Food scraps in landfills
& incinerators
generate pollutants
like methane...



...but food scraps
converted into
compost actually
sequester
carbon.

4 TO PROTECT THE CLIMATE



Composting Protects the Climate

Food scraps in landfills generate methane, a greenhouse gas with a global warming potential 84x more potent than CO_2 , in the short term.

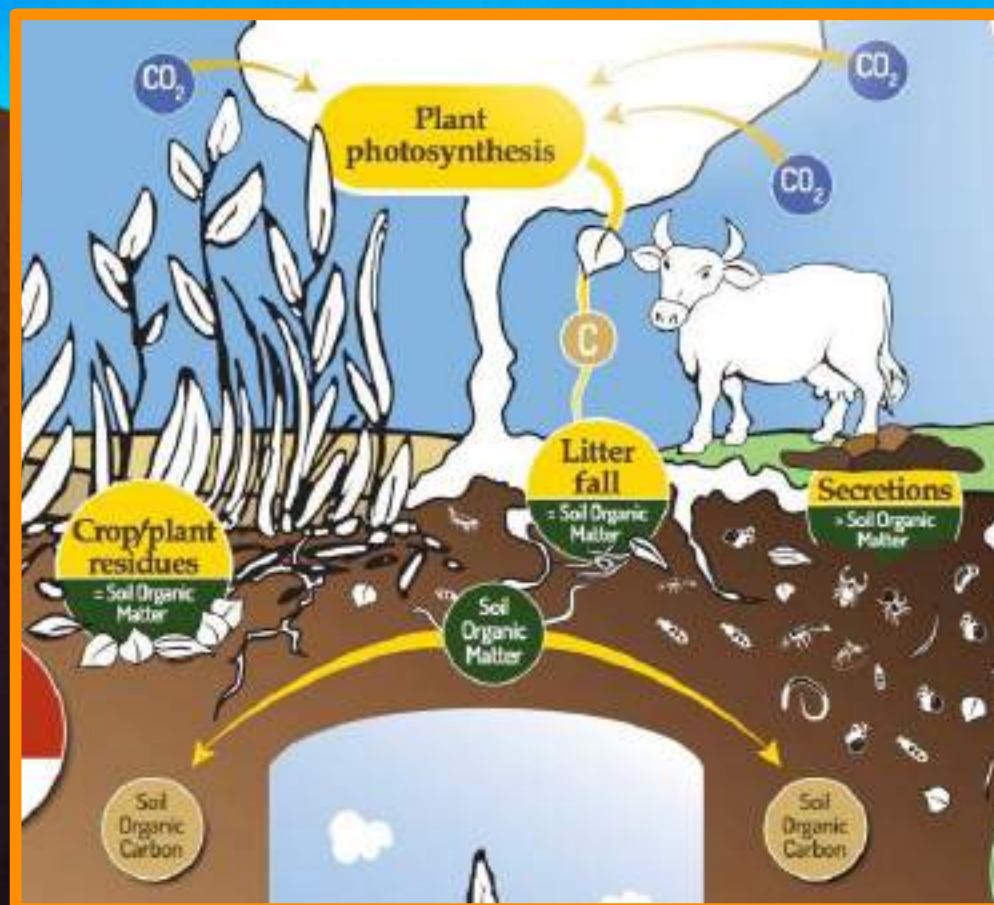
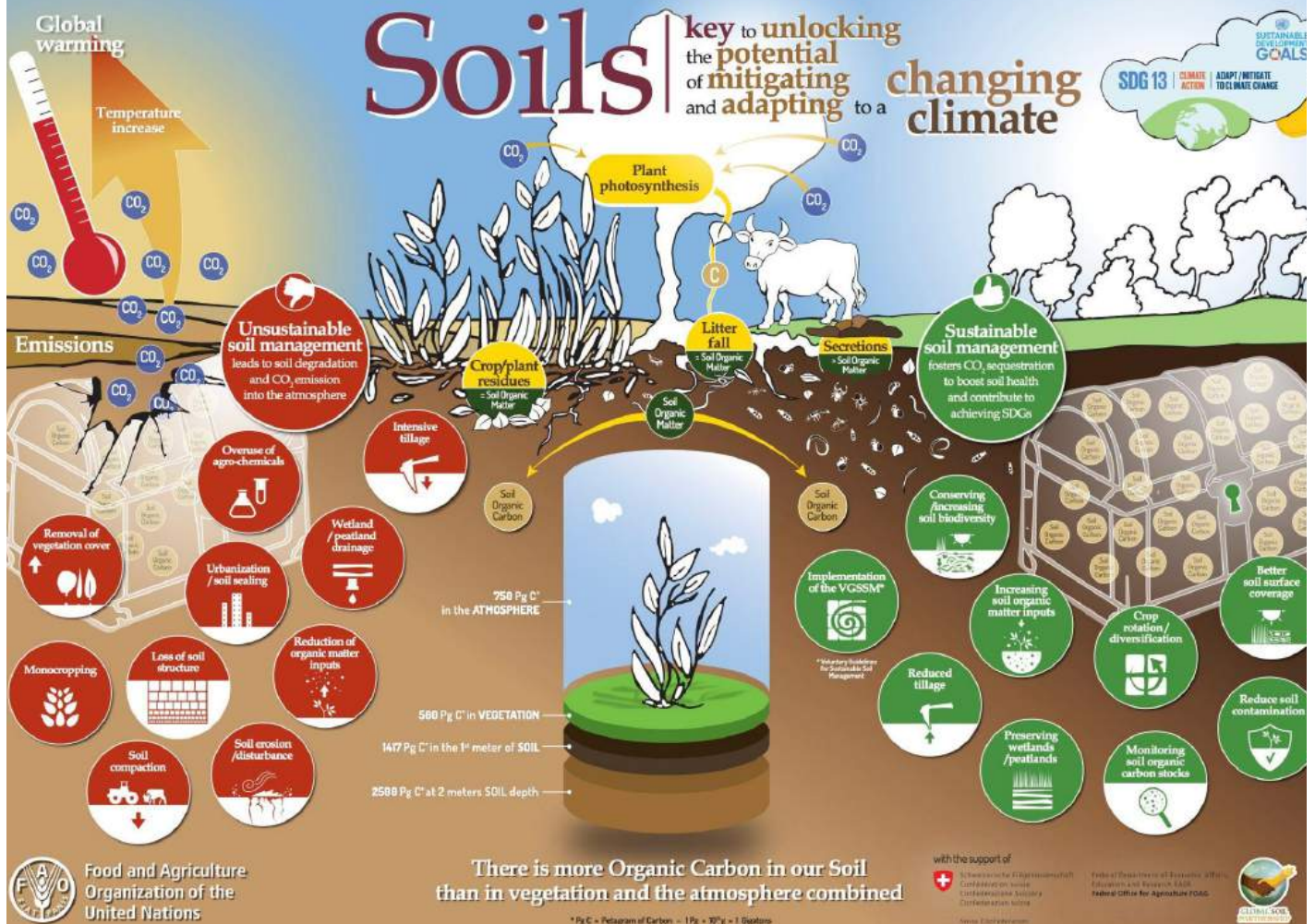


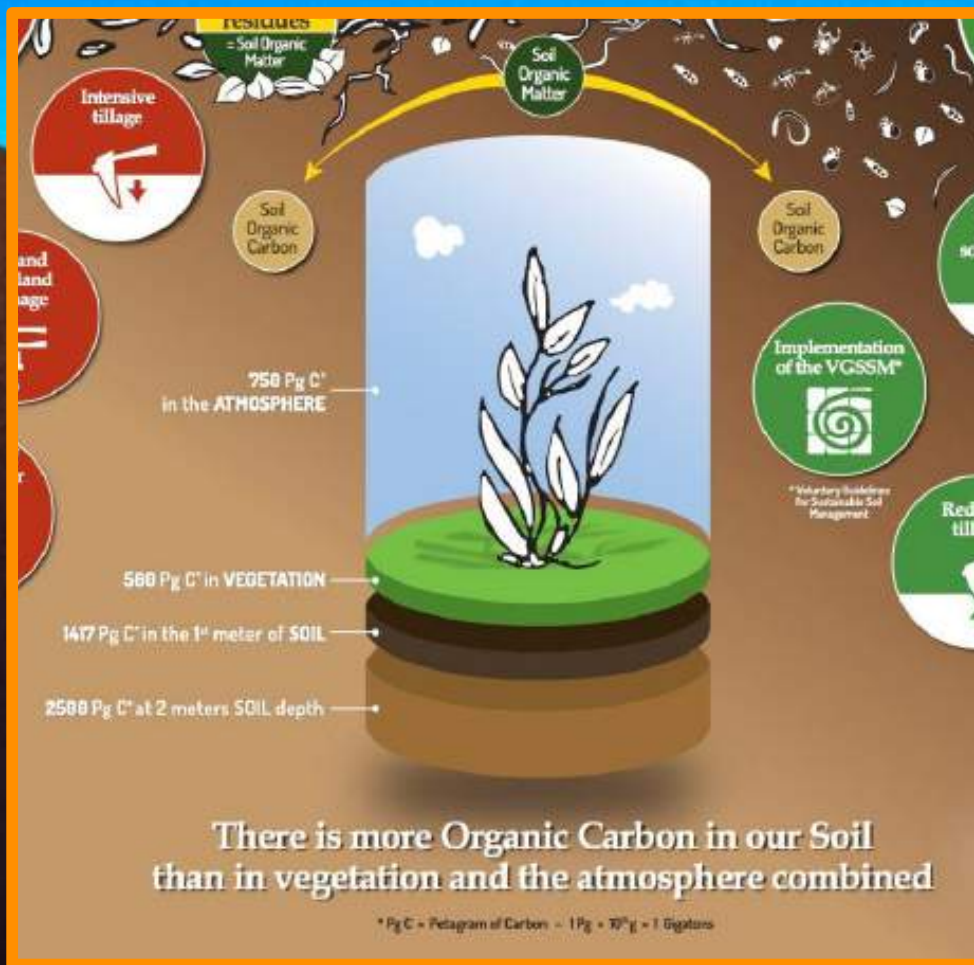
...but when converted into
compost and applied to the land,
compost sequesters carbon.

One research project found that $\frac{1}{2}$ inch of compost applied to rangeland sequestered the equivalent of 1 metric ton of CO_2e /hectare over three years.

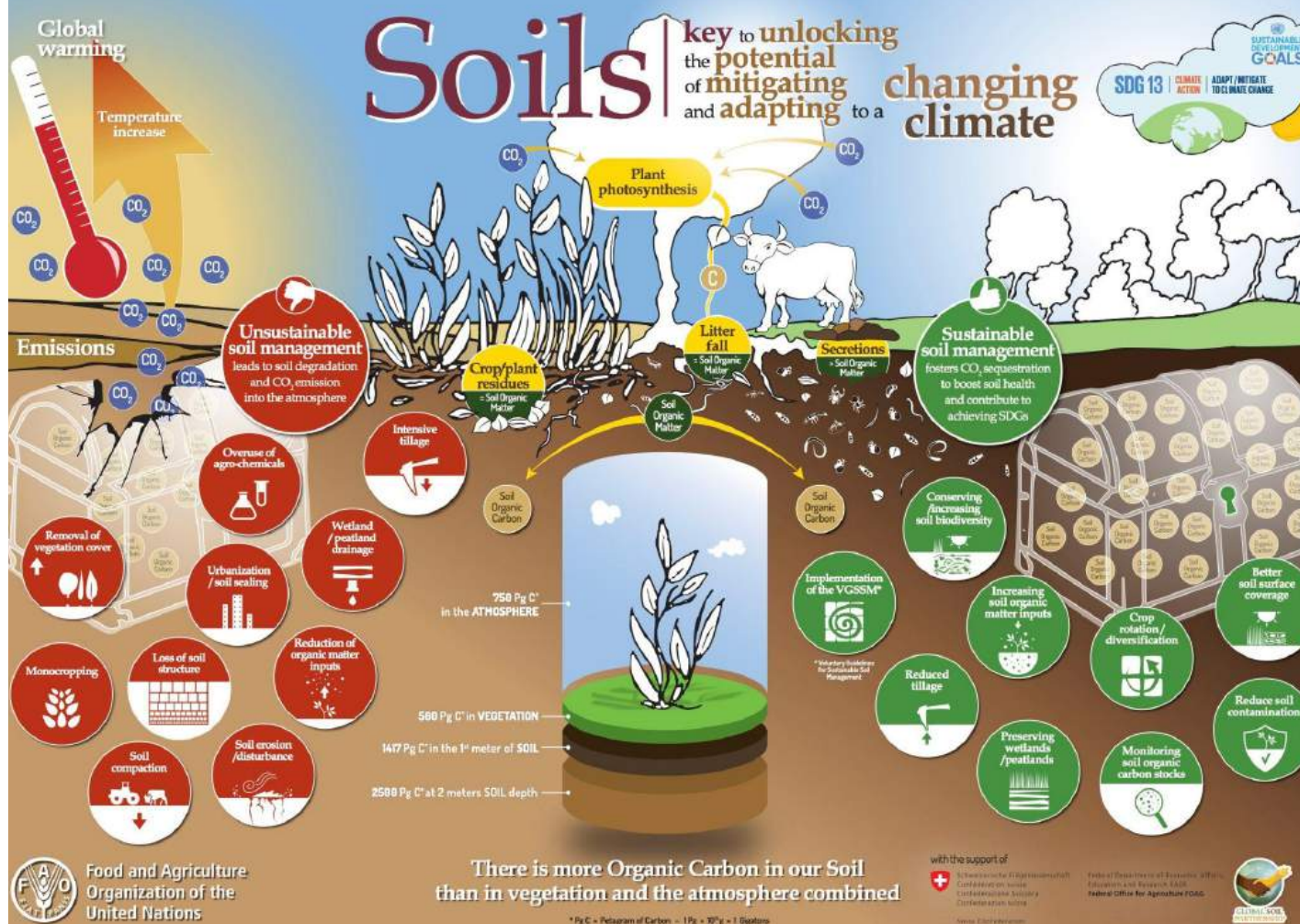
This level of sequestration on half of California's rangeland would offset 42 million metric tons of CO_2e , which is equal to the annual greenhouse emissions from California's commercial and residential energy sectors.

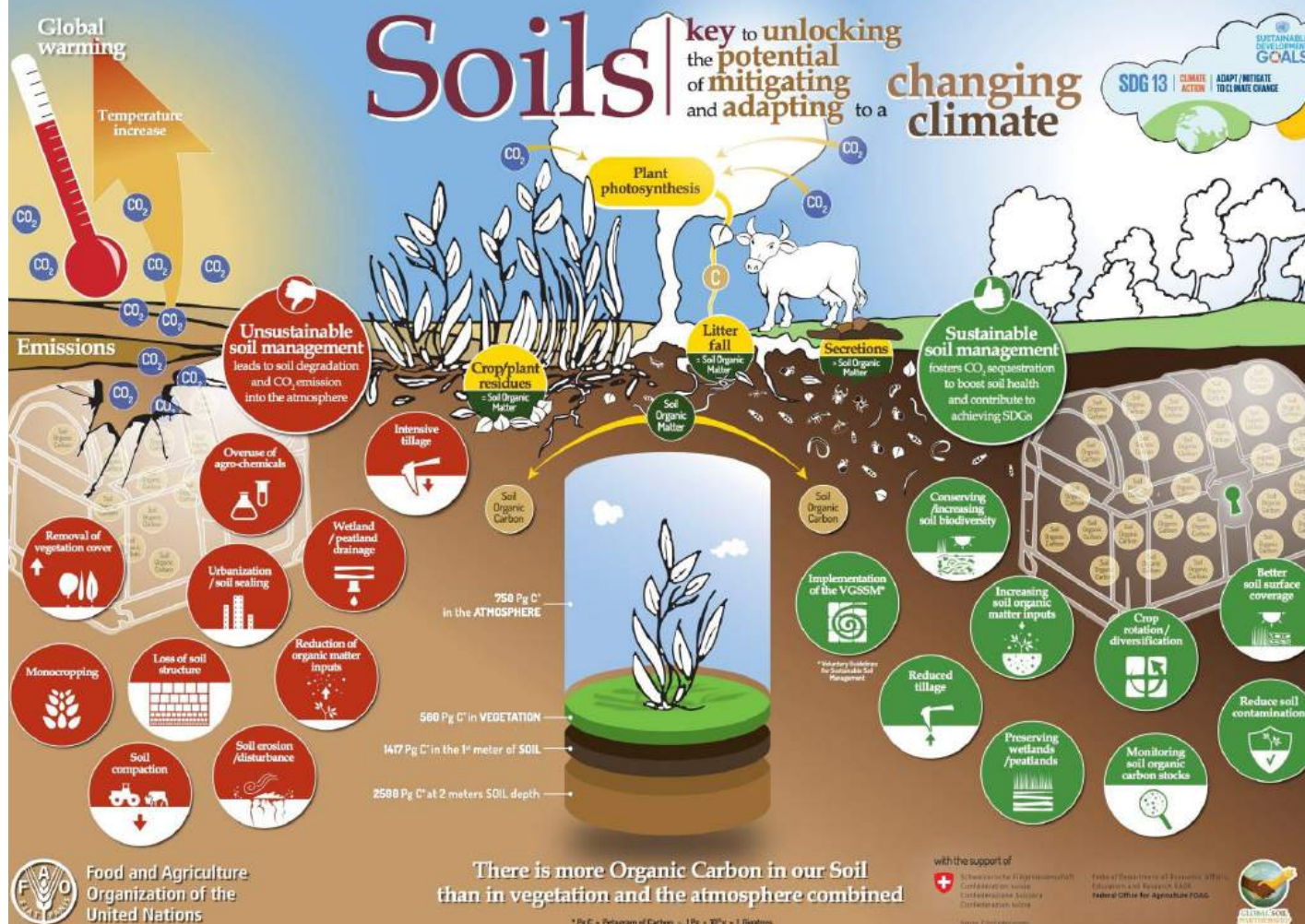
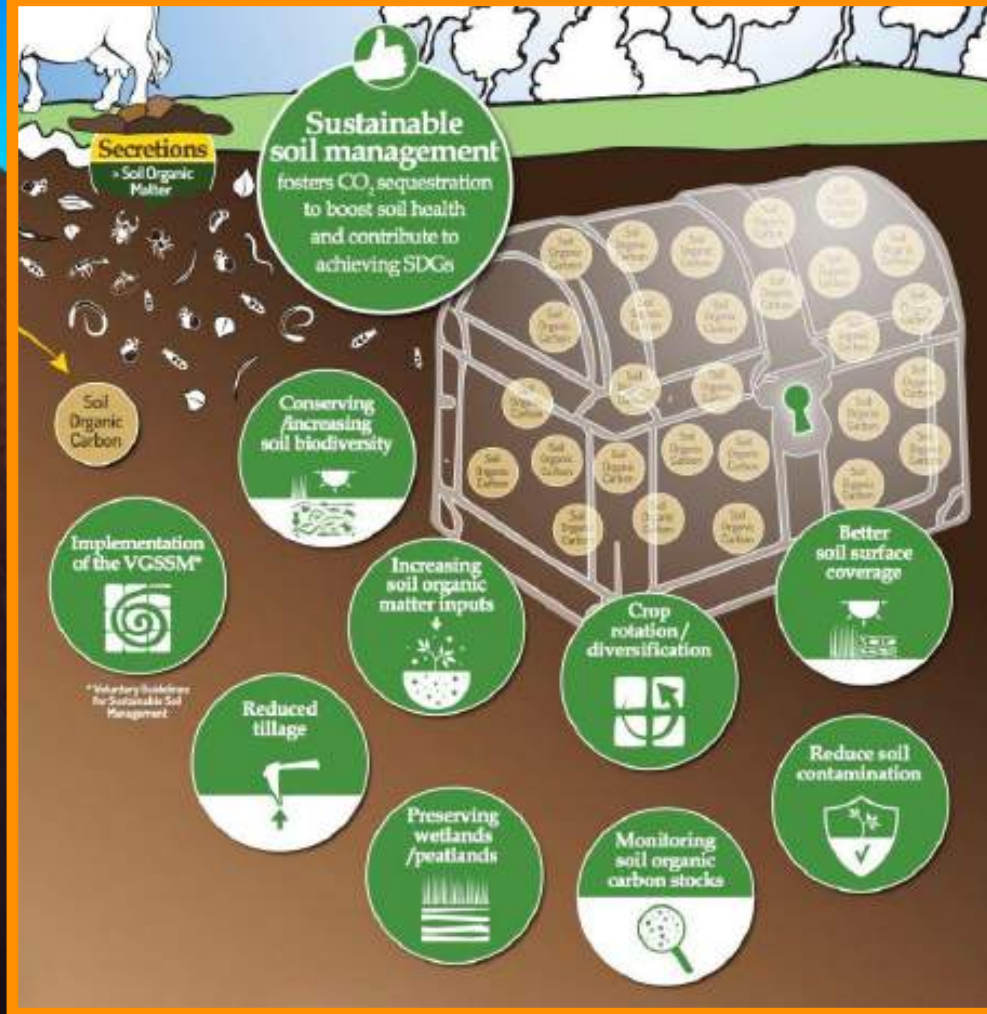


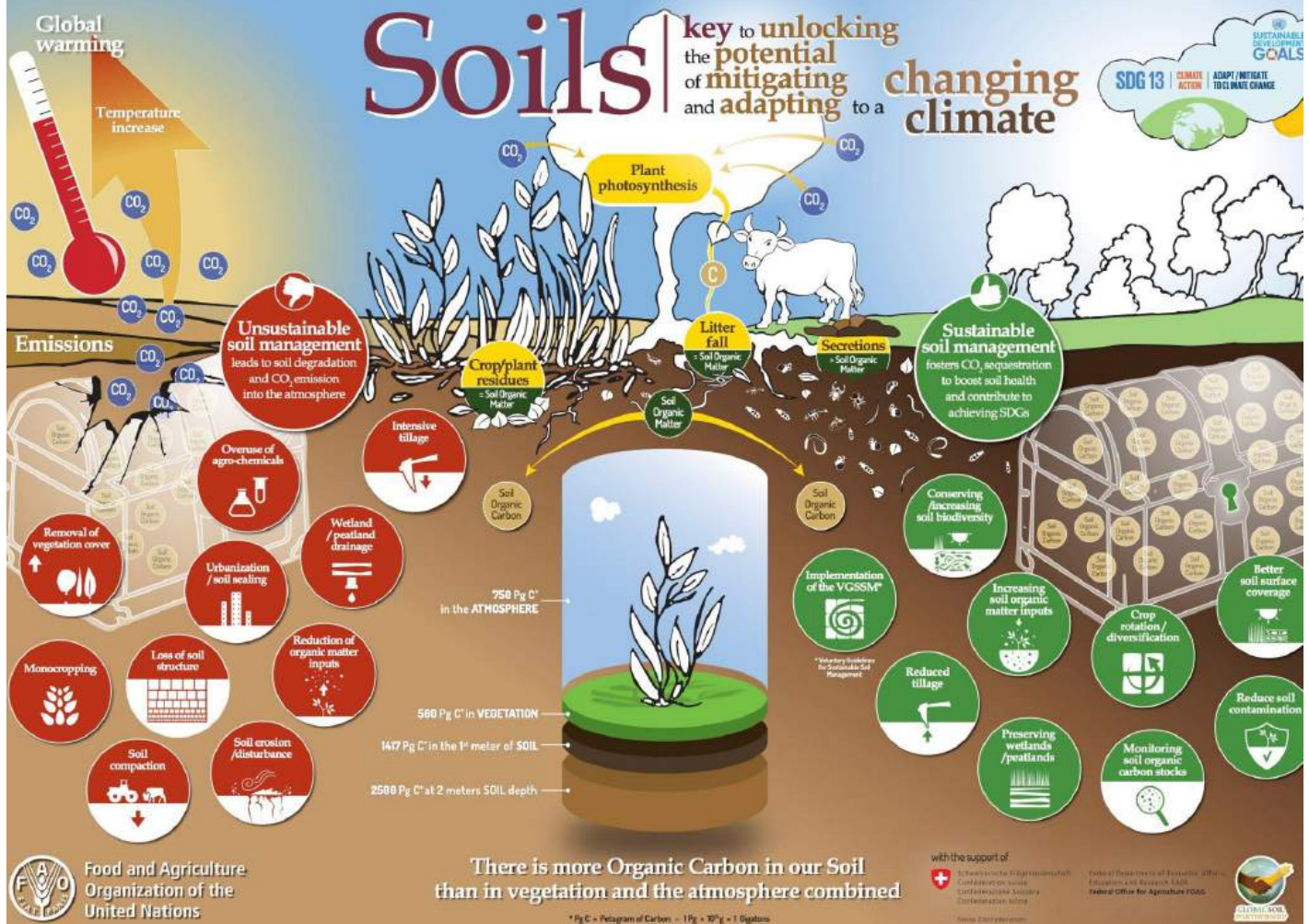
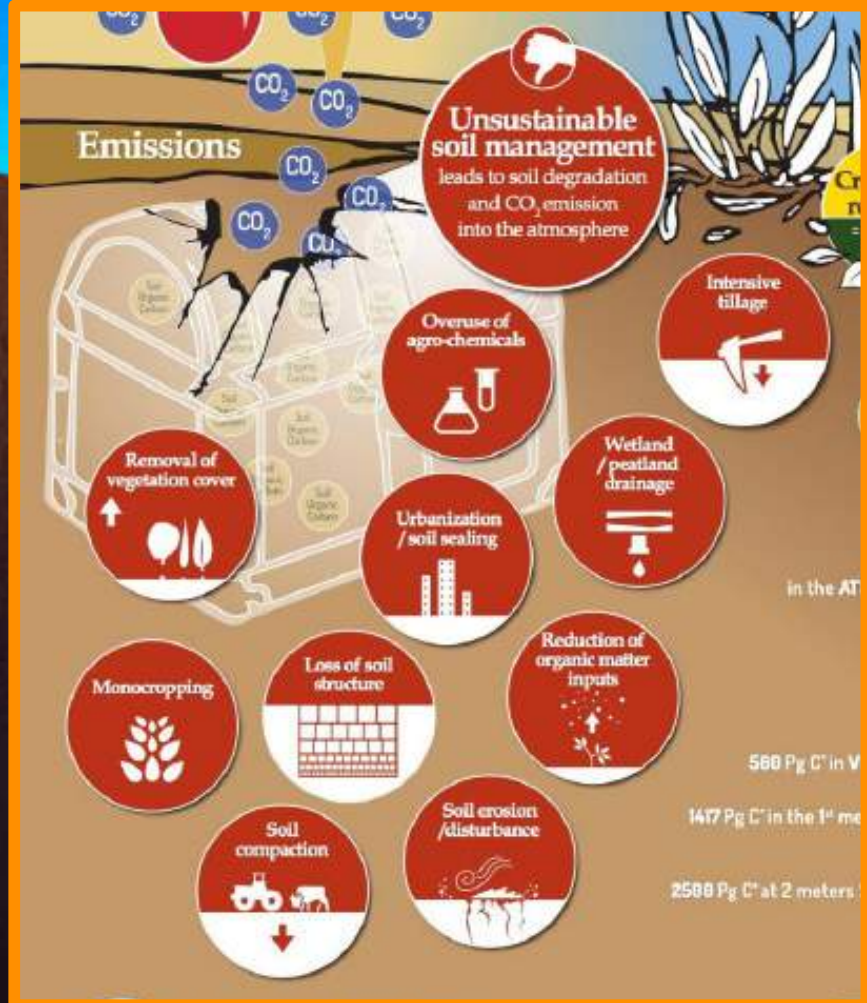




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Compost = Climate & Soil Protection Strategy



Photo: Marin Carbon Project

- Prevents landfill methane emissions
- Adds carbon to soil
- Improves soil's ability to store carbon
- Improves plant growth, and thus carbon sequestration
- Reduces need for energy-intensive fertilizers, pesticides, fungicides
- Reduces energy use for irrigation



The *What* & *Why* of Composting

- ✓ Defined key composting related terms
- ✓ Introduced the composting process
- ✓ Discussed benefits of composting
- ✓ Hierarchy to Reduce Food Waste & Grow Community



Image: @LACompost



Composting Systems & the Role of Community Composting

- ✓ Composting Scales & Systems
- ✓ Define Community Composting
- ✓ Examples of Community Composting
- ✓ The Importance of Training